GREEN INFRASTRUCTURE FRAMEWORK (GIF) - GUIDING PRINCIPLES

EAST SWINDON

THE DELIVERY OF INTEGRATED BIODIVERSITY, LANDSCAPE & RECREATIONAL RESOURCES

fpcr

October 2013

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Site Location

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masterplanning ■ environmental assessment ■ landscape design ■ urban design ■ ecology ■ architecture ■ arboriculture

S:\1796\Green Infrastructure Strategy\1796-L-01 Green Infrastructure Strategy.indd

INTRODUCTION

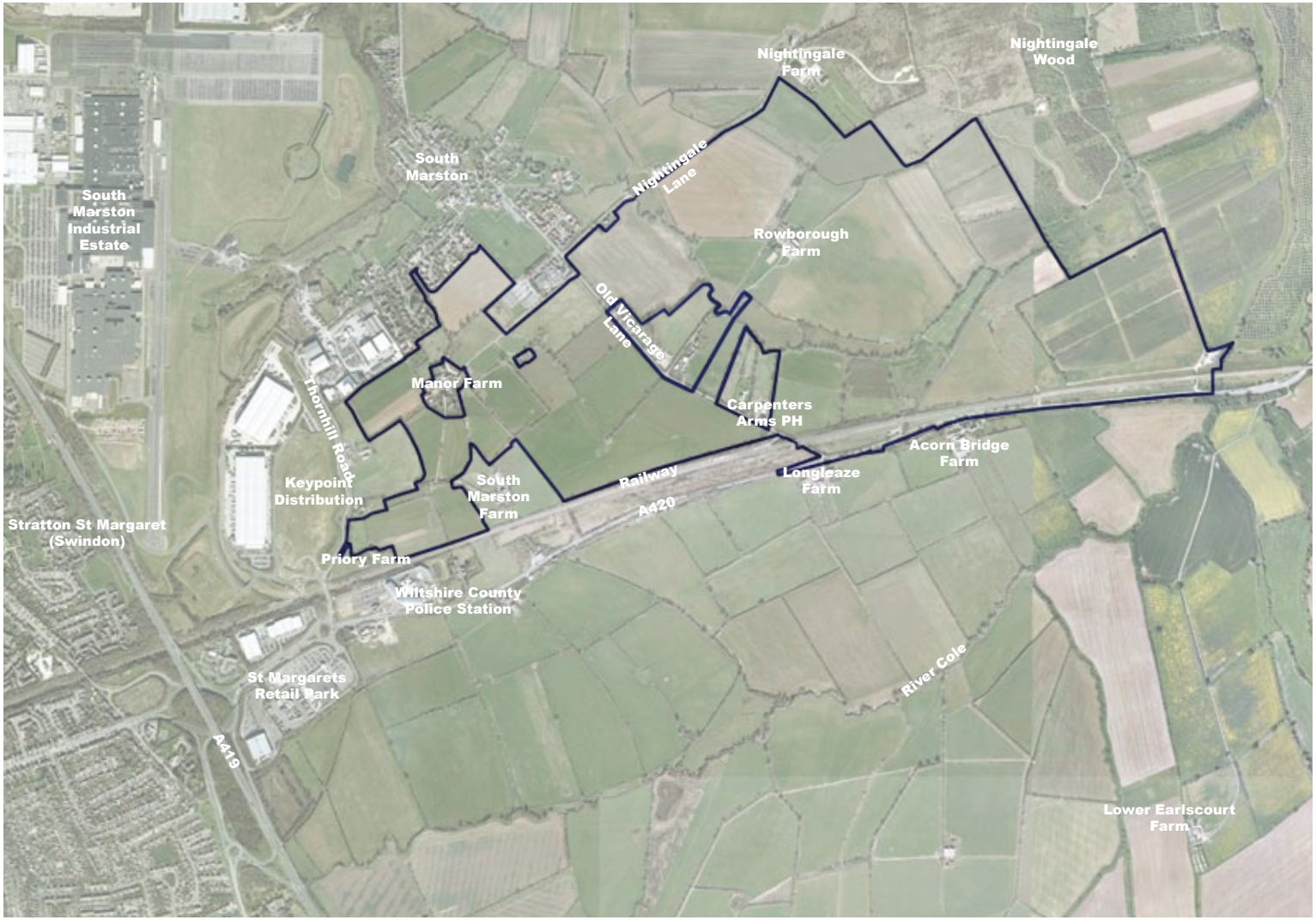
The proposed development at East Swindon seeks to deliver a new sustainable community of housing, primary education, retail, community facilities, sports land and open space.

The proposed Green Infrastructure Framework (GIF) delivers around 70ha (over 40% of the site) of multifunctional greenspace, providing long term biodiversity, landscape and recreational benefits.

The Green Infrastructure Framework (GIF) for the new community has been prepared in the context of a thorough and detailed understanding of the site and its context, and within a framework of key policy and environmental design guidance.

The existing landscape and biodiversity resource of the site has been considered by the assessment, masterplanning and design process.

This document sets out the overarching guiding principles for East Swindon's GIF. It provides a mechanism in which to secure a high quality GIF and it is anticipated that the document will be used as a template for the detailed landscape design stage.



Aerial Photograph

ENVIRONMENTAL DESIGN & PLANNING GUIDANCE

NATIONAL CONTEXT

The National Planning Policy Framework focuses on the positive delivery of sustainable development. It does not introduce new environmental assessment or design criteria, but instead builds upon the previous robust guidance found in Planning Policy documents to promote an holistic approach which balances overall benefits arising from development with inevitable impacts which arise and the ability to successfully mitigate harm.

Under Core Principles, the drive is for high quality integrated design which raises standards. The "intrinsic character" of the landscape setting for any site is to be recognised, and schemes should contribute to the conservation and enhancement of the natural environment. The objective is to minimise impacts on biodiversity and provide net gains overall where possible, establishing coherent, resilient ecological networks. The NPPF counsels local planning authorities to adopt a strategic approach, planning positively for the creation and management of networks of biodiversity and green infrastructure (Para 114).

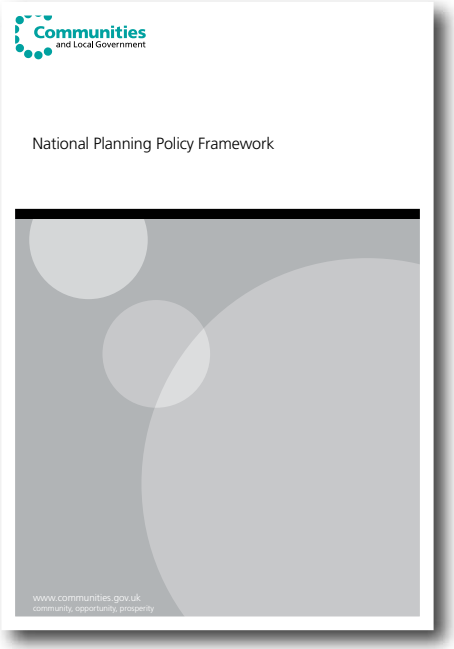
The Government now places very great importance on the delivery of sustainable and co-ordinated green infrastructure (GI), as a key component of its sustainable development agenda. Natural England and Sport England, the Government`s principal agencies tasked with promoting landscape, biodiversity and healthy living, fully endorse this all embracing approach.

"Green Infrastructure Guidance" articulates Natural England`s (NE) position in relation to GI planning and delivery, which is seen as a key part in adapting to and mitigating climate change and to delivering multiple environmental functions. The NE guidance recognises the importance of "Multifunctionality" to GI. Of particular note is the contribution that high quality GI can play to the delivery of a number of NE`s strategic outcomes as listed on page 13 of the document. These include:

"Well planned Green Infrastructure encompassing new and enhanced sites and habitats:

- *contributes to high quality and accessible landscapes benefiting people and wildlife;*
- *plays an essential role in maintaining and enhancing the health of the natural environment and its ability to provide a wealth of 'ecosystem services';*
- *increases ecological connectivity to overcome habitat fragmentation and*
- *increase the ability of the natural environment to adapt to climate change"*

The creation and enhancement of green infrastructure helps to:



- *create attractive and accessible places for people to enjoy direct and regular contact with the natural environment;*
- *strengthen links between urban areas and their surrounding countryside, and bring the natural world into every neighbourhood, with benefits for individual and community health and well-being."*

"Nature Nearby – Accessible Natural Greenspace Guidance" is another NE guidance document focussed upon the delivery of high quality and accessible green spaces close to where people live and work and upon the connection of people with the natural environment. The three underlying principles are:

- *Improving access to green spaces.*
- *Improving naturalness of green spaces.*
- *Improving connectivity with green space.*

In addition to the NE guidance documents, a series of other policy and design guidance based document have informed the over arching principles for the GIF. These include:

- *Green Infrastructure, An integrated approach to land use, Landscape Institute Position Statement, March 2013*
- *Active Design – Sport England*
- *A Green Infrastructure Strategy For Swindon 2010 - 2026*
- *Draft Swindon Borough Local Plan 2026*
- *Swindon Borough Council: Landscape*

LOCAL CONTEXT

DRAFT BOROUGH SWINDON LOCAL PLAN

New Eastern Villages and South Marston

The site falls within the allocation that forms part of the proposed expansion of South Marston (Policy RA3). This forms one of the New Eastern Villages as part of Policy NC3.

Policy NC3 states the development shall provide:

- An extensive green infrastructure network that maximises opportunities for habitat connectivity and enhanced biodiversity including extending the River Cole green infrastructure corridor and connecting with Nightingale Wood;

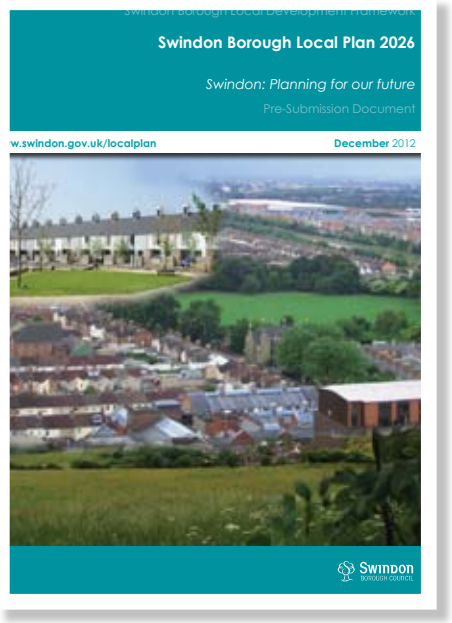
- Sports and leisure facilities.

In summary, and in relation to landscape matters, Policy RA3 states:

"b. Development at South Marston shall:

- Ensure it protects the character of the existing village:
- Provide green infrastructure in accordance with Policy EN1 proportionate to the scale of expansion;
- Provide an extended recreation ground to include the field to the south west of the current recreation ground with recreational facilities of an appropriate type and scale;
- Protect historical landscape features, archaeological assets and existing green infrastructure;

c. To ensure non-coalescence of South Marston with Swindon and the other proposed Eastern Villages and the land between the expanded village and the railway to the south shall remain part of the countryside…"



LOCAL CONTEXT

Natural and Built Environment

Turning to general landscape matters, Theme 7 of the Swindon Borough Local Plan 2026: Submission Dcoument (2013) sets out polices to:

"protect, enhance and manage the natural and built environment; ensure that the appropriate provisions are in place to deliver green open space as part of new development".

Policy EN1 is concerned with Green Infrastructure (GI). The Policy states:

"a. Development shall protect and enhance green infrastructure and assets as identified in Appendix 5. This includes the requirement that development must provide for the protection and integration of visually or ecologically important existing trees, hedges and woodlands.

b. Development shall design green infrastructure to maximise its connections and functions and ensure the sustainable maintenance and management of it."

Paragraphs 4.324 and 4.325 go on to say: "...

Swindon's green infrastructure will continue to play a key role in Swindon's sustainable growth and regeneration and it will sit at the heart of a far-reaching network of interconnected green spaces that:

- Connect and enrich biodiversity habitats;
- Improve access, recreation and tourism opportunities;
- Contribute to the health and well being of the residents of Swindon;
- Enhance landscape character and respect local distinctiveness;
- Help meet the challenges of climate change; and;
- Realise the potential of historical and cultural heritage features to contribute to local identity and sense of place.

The Swindon Green Infrastructure (GI) Strategy provides a comprehensive plan for the protection of existing and the creation of new green infrastructure in the borough."

Draft Eastern Village Supplementary Planning Document (July 2003) and South Marston Draft Supplementary Planning Document (June 2013)

Supplementary Planning Documents are in the course of preparation for the eastern Villages and for South Marston and have recently been the subject of consultation. They provide further more detailed guidance.

The Eastern Villages Draft Supplementary Planning Document (2013) sets out the Eastern Villages Vision, which is to create well-designed places where people want to live, work and spend time. Amongst the key principles, the development will:

- sensitively and positively respond to the existing landscape context, natural and historic assets and the character and identity of the surrounding villages as well as enhancing biodiversity and green infrastructure;

- comprise new distinct villages with individual identities and characters linked together by green spaces that integrate with the existing urban area and wider landscape setting.

LOCAL CONTEXT

A Green Infrastructure Strategy for Swindon 2010-2026

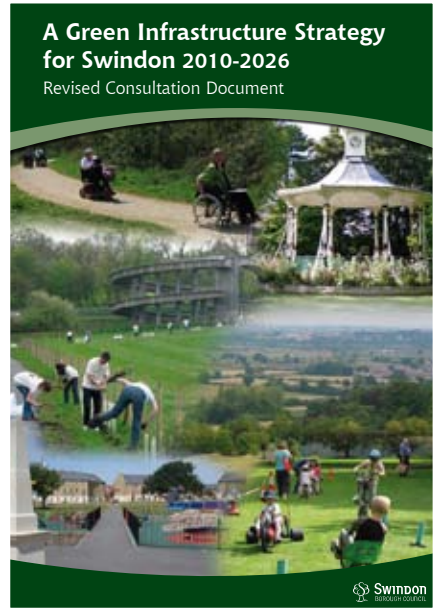
The Borough Council have prepared a Green Infrastructure (GI) Strategy. Its aims are as follows to:

- "Prioritise the planning development of and investment in green infrastructure in Swindon to 2026.
- Preset a shared vision for the development of a strategic green infrastructure network across the Borough of Swindon and reaching into neighbouring areas.
- Highlights by means in which the organisations, communities and partnerships, can work to create and sustain a fit for purpose green infrastructure network across the area.
- Consolidate the essential role green infrastructure will play in the sustainable development of Swindon."

The GI strategy will be used to:

- "Provide the basis for the coordinated approach to the creation and sustained management of green infrastructure across Swindon and its neighbouring authorities.
- Support and guide the production of local development plan documents, particularly a Green Infrastructure Supplementary Planning Document for Swindon and master plans associated with major development areas.
- Provide a framework for the development of other localised green infrastructure plans.
- Act as a reference document to other key strategies and such as community plans, sustainability strategies, local transport plans, and climate change action plans.
- Act as a basis for a rolling 3 year GI delivery plan."

Section 4: Proposals for enhanced Green Infrastructure across Swindon includes a vision and a series of aims and objectives. This includes proposed GI networks across Swindon that are based upon strategic corridors, sub regional corridors and local clusters. The Eastern Development Area is identified as a "local GI cluster".



DEVELOPMENT PROPOSALS

INTRODUCTION

The proposed development comprises the following:

A sustainable urban extension to the east of Swindon and north of the A420 of up to 2,380 dwellings together with a mixed use local centre and mixed use areas:

- including A1 retail up to 2,500 sq metres, services (A2), restaurants, pubs and takeaways (A3, A4, A5), business uses (B1), up to 2,500 sq metres, residential;
- community uses (D1);
- sheltered accommodation;
- primary school;
- green infrastructure including formal (including playing fields) and informal open space, landscape and amenity space; changing and sports facilities (including D2);
- sustainable drainage systems; highway, cycle and pedestrian routes; car parking; infrastructure (including utilities);
- engineering works including ground remodelling; demolition, site reclamation and removal of structures;
- the formation of new accesses from the A420, Old Vicarage Lane and Thornhill Road.

THE INDICATIVE MASTERPLAN

SITE CHARACTER

Context

In broad terms, the site forms a parcel of agricultural land lying between the small village of South Marston to the north and the A420 and the Great Western railway line to the south.

Although predominantly an agricultural landscape, the site lies close to the urban edge of Swindon and there are a number of urbanising elements that lie near to the site, or in close proximity which have a strong urban influence on the landscape. This includes, amongst others, the A419 (Swindon by-pass), A419-A420 interchange and large buildings at St Margaret's Retail Park to the west; the extensive South Marston Industrial Estate and Keypoint Distribution Centre to the north and west which includes a number of large employment buildings and distribution warehouses, and the railway and A420 which run parallel with one another to the south.

The site boundary is defined by a combination of the curtilage of farms and properties, hedgelines, public rights of way, roads and by the edge of South Marston.

South Marston village (together with Nightingale Lane) form the site's northern boundary. The southern edge of the village is defined by residential properties, South Marston primary school the Mercure South Marston Hotel and Leisure Club, as well as an industrial/employment area off Thornhill Road.

Residential properties that border, or lie close to the site, are largely modern late 20th century in character and these include properties at Ash Gardens, Manor Park, Rawlings Close, Bell Gardens, and Yew Tree Gardens. In general, the older part of South Marston is concentrated around the Church (St Mary Magdalene) at the northern end of Old Vicarage Lane, and around the junction with Chapel Lane and Thornhill Road-Highworth Road.

Nightingale Lane also defines the site's northern boundary. This narrow lane extends in a north-easterly direction from South Marston towards Nightingale Wood, which Nightingale Wood is publically accessible woodland of broadleaved and coniferous trees that that was planted in the 1990s. The site's eastern boundary is defined by the Wood and by hedgerows.

The River Cole lies to the east of Nightingale wood, set within a continuous belt of woodland that stretches from Sevenhampton to the north and the A420 to south. To the east of the River Cole are the larger settlements of Shrivenham and Watchfield, and the smaller villages of Lower Bourton and Bourton.

The site's western boundary is defined by a series of hedgerows close to Thornhill Road. Further to the west and north of South Marston is the industrial and employment area of South Marston Industrial Estate, which is accessed from the A361. The area includes a number of large scale employment buildings and distribution warehouses and this includes the Honda complex, G-Park (DHL and B&Q Distribution Centre), and the recently constructed buildings at Keypoint Distribution and Keypoint Railfreight Interchange.

To the north of South Marston are the villages of Stanton Fitzwarren and Sevenhampton.

The site's southern boundary is defined by the railway line which for most of its course alongside the site is on embankment, and the A420 which lies immediate south of the railway line following a similar alignment. Wiltshire County Police HQ, St Margaret's Retail Park and a Sainsbury's retail store lie to the west close to the A420 -A419 (Swindon by-pass) interchange.

The landscape to the south of the A420 is characterised by the lower lying agricultural landscape of the River Cole. Fields and hedges are interspersed with scattered farms and properties which include Lotmead Farm, Mount Pleasant Farm and Lower Earlscourt Farm.

The villages of Wanborough, Hinton Parva and Bishopstone lie on the edge of the North Wessex Downs scarp above the River Cole valley and the urban area of Swindon. There are extensive views from the escarpment across the valley and of Swindon.

A number of farms and properties lie near to the site. This includes Priory Farm, Manor Farm, Acorn Bridge Farm, Longleaze Farm, Marston Farm, South Marston Farm, Home Farm, Rowborough Farm and Red House. As well as residential properties on the southern edge of South Marston, there are also a number of properties on Old Vicarage Lane as it heads southwards towards the A420.

Designations

The site is not subject to any statutory landscape quality designations, nor is the local landscape. Within the wider study area is the North Wessex Downs AONB to the south, which lies around 4km from the site's closest point, and the Cotswolds AONB around which lies to the north of Swindon, approximately 13km from the site at its closest point.

Conservation Areas are located in Lower and Upper Wanborough and Hinton Parva on the North Wessex Downs escarpment, Stanton Fitzwarren and Sevenhampton to the north, and within Stratton St Margaret (in Swindon) to the west. In close proximity to the site are the farmhouses of Priory Farm, Manor Farm, and Nightingale Farm which are Grade II Listed, as is the property of The Red House.

Land Use

The site is currently in agricultural use and comprises arable fields with some improved pasture for cattle grazing. The field pattern is typical of the wider landscape with fields being generally irregular in shape and predominantly of small scale, although there are some bigger fields to the east near Rowborough Farm

Hedgerows and ditches define the field pattern. Many of the hedges are tall and thick containing a number of mature trees, although some are variable in quality, largely due to current management practices. As well as some wet ditches associated with hedgelines, there are some small field ponds within the site. Broad leaved plantation woodland occurs in the form of recently planted, willow dominated woodland. Some coniferous planting also occurs.

The narrow South Marston Brook (a tributary of the River Cole) runs through the eastern part of the site. It is heavily shaded for much of its length by adjacent trees and scrub.

There are a number of Public Rights of Way (PROW) that cross the site, mostly in a north-south direction between South Marston and the A120. A Public Bridleway heads into Nightingale Wood via Rowborough Farm.

Topography

The main part of the site is relatively flat in character. The site varies between 100-90m Above Ordnance Datum with the higher part of the site at Nightingale Wood to the east. The site gently falls in a south easterly towards the Marston Brook and the River Cole.

The landform begins to gently rise to the north of South Marston at Stanton Fitzwarren (124m AOD) and Reddown Hill (135m AOD), before reaching higher land of the Cotswolds AONB to the north. To the south of the site there are some small occasional rises at Mouth Pleasant Farm (103m AOD) and at Bourton (105m AOD), but in the main the landscape is characterised by the wide and generally flat Cole Valley (90-95m AOD). Steeply rising slopes occur on the edge of the North Wessex Downs Scarp with Upper Wanborough at 140m AOD. Charlbury Hill, near Hinton Parva reaches 253m AOD.

Biodiversity

A range of habitats are present within the site and are typically heavily influenced by the current agricultural management of the site. The site is dominated by arable fields and improved pasture divided by hedgerows. There are occasional species-poor, semi-improved grasslands present in association with field boundaries and areas of less intensive management. Mature trees are frequently associated with hedgerows.

Significant parts of the site are planted with willow coppice, which, it is understood, was planted as a source of biofuel, although remains un-harvested.

Semi-improved grassland is generally highly localised within the site and persists where intensive agricultural management is reduced. All are species–poor in nature and where ungrazed, such as adjacent to Nightingale Lane, are dominated by a range of common coarse grasses.

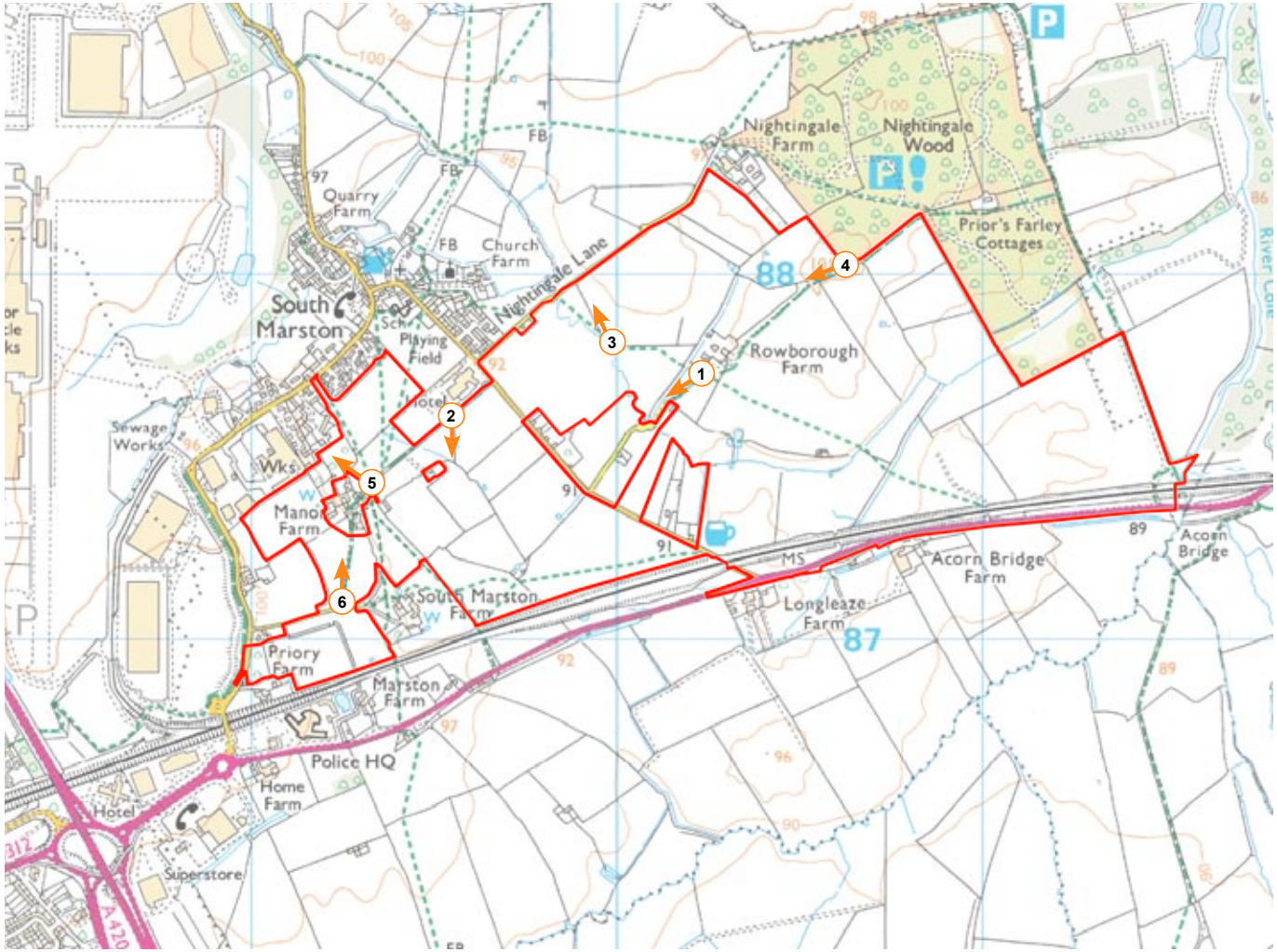
Small patches of marshy grassland occur to the south of South Marston, where damp depressions/ ditches occur. A suite of grassland species typical of water-logged conditions were prevalent.

Tall ruderal vegetation is located frequently along hedgerows bases as is typical where arable and improved grassland predominate.

Representative Viewpoints

Site Boundary

Viewpoint Location



Representative Viewpoints

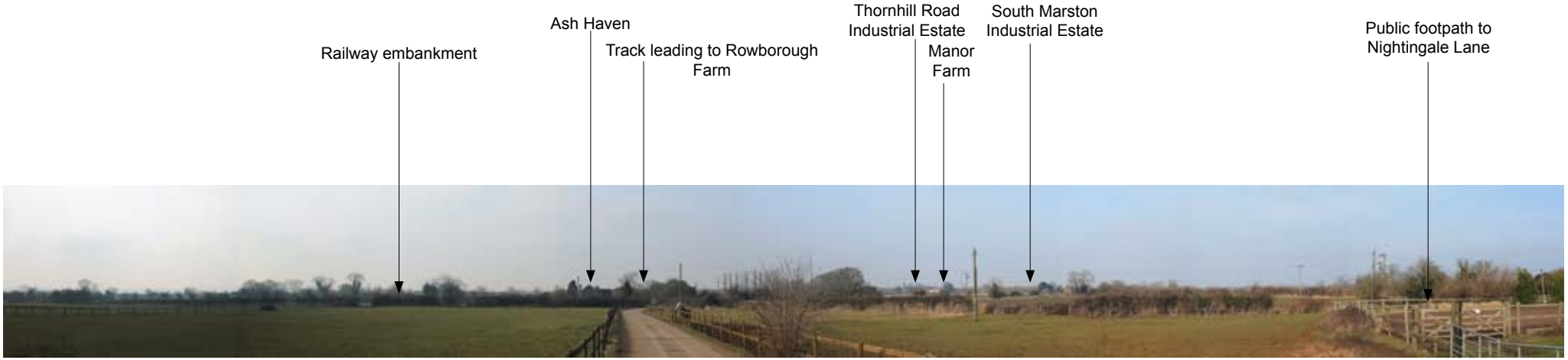


PHOTO VIEWPOINT 1: View looking west from public bridleway - public footpath near Rowborough Farm.

PHOTO VIEWPOINT 2: View looking west and south from public bridleway near Mercure Hotel.

Representative Viewpoints

PHOTO VIEWPOINT 3: View looking north and east from public footpath near South Marston Brook.

PHOTO VIEWPOINT 4: View looking west from public bridleway near Nightingale Farm.

Representative Viewpoints

PHOTO VIEWPOINT 5: View looking from public bridleway, near Manor Farm.

PHOTO VIEWPOINT 6: Track leading to Manor Farm.

GREEN INFRASTRUCTURE FRAMEWORK

The provision and approach to GI is in accordance with **A Green Infrastructure Strategy for Swindon 2010-2026** and the iniatitives of the Great Western Community Forest. It also accords with established guidance from Natural England and Sport England in terms of accessibility to greenspace for new communities and delivering multifunctional landscapes that offer recreation and biodiversity gains.

The Development Framework Plan and the location of development parcels has been shaped by the principal natural assets such as South Marston Brook, the network of hedgerows and mature trees, and rights of way. In addition to this, the Development Framework Plan has been carefully devised to take into account the context and setting of nearby residential properties within South Marston, as well as surrounding farmsteads such as South Marston Farm and Manor Farm.

The Development Framework Plan responds to the site's landscape character by conserving features of landscape value within the site (brook, important hedgerows and mature trees). These are conserved within areas or corridors of new greenspace and will be strengthened by new habitat creation.

The proposed development will incorporate strategic GI proposals. These will comprise the conservation of existing woodland, hedges and trees, reinforced by significant new woodland, tree and shrub planting; the creation and conservation of landscape biodiversity corridors throughout the development; the provision of new mixed habitats (including wetland areas as part or the sustainable drainage strategy) to satisfy biodiversity objectives. The GI also includes extensive areas of accessible greenspace for active sport, natural play, walking and cycling for a healthy living.

There are 10 guiding objectives for the development's GIF:

1) To embrace the principles and aspirations that are set out within A Green Infrastructure Strategy for Swindon 2010-2026, the Great Western Community Forest, and Natural England's Green Infrastructure Guidance.

2) To ensure that the GIF framework is multifunctional in its design so that it delivers long term landscape, biodiversity, recreation and sustainability benefits.

3) To minimise impacts on landscape character and biodiversity by retaining existing landscape features of value, such as important hedgerows and mature trees, and to strengthen these habitats with new broadleaved planting and landscape habitats to provide net gains overall.

4) To deliver a varied GIF framework which includes existing habitats, as well as the creation of swales basins, ponds and wetland areas; species rich hedgerows; broadleaved woodland; native trees and shrub planting; and grassland meadows.

5) To develop a mix of species (to be finalised at the detailed stage) which will be based upon an appropriate range of native species that can be found within the local and wider landscape of the Midvale Ridge Character Area.

6) To ensure that the GIF framework is interconnected so that it to provides greenspace and wildlife corridors that permeate within and around the development, with connections to South Marston Brook and Nightingale Wood.

7) To create a GIF that is publically accessible so that all new residents can easily access an area of greenspace for walking and cycling, as well as play, sport and informal recreation.

8) To deliver new sports provision, equipped play, pocket parks and large areas of areas of accessible open space for active sport and play and informal recreation and healthy active living.

9) Retaining the site's rights of way (diverting where necessary) and creating new multipurpose recreational routes routes within the GIF for walking and cycling, that provide access around the proposed development as well as delivering improved access into the wider landscape.

10) To minimise the impact of new development on the existing community of South Marston and surrounding properties that lie in close proximity through a well-considered masterplan and GIF framework.

GREEN INFRASTRUCTURE OVERVIEW

KEY

- Development Area
- Green Infrastructure

LANDSCAPE ZONES

The GIF framework equates to some 70ha or c40% of the site. It includes the following components.

1) The creation of a large multifunctional landscape habitat within the Proposed Development. Although on 'the ground' this will be observed a single extensive and connected area of greenspace offering a variety of habitsts it is, for ease, separated into three areas to aid explanation See Green Infrastructure Framework (GIF) - Guiding Principles, for locations)

- Zone 1: South Marston Park. This is located to the west of Old Vicarage Lane. It encompasses retained trees, hedges and rights of way, together with new habitat creation in the form of new broadleaved planting of woodland copses, hedges and trees and grassland meadows and SUDS (basins and swales). The park will include new sports pitches (alongside the existing recreation ground), and areas of equipped play.

- Zone 2: Rowborough Park. This lies to the east of Old Vicarage Lane and is broadly defined by South Marston Brook. It encompasses the Brook and its associated tree cover and floodplain, as well as retained trees, hedgerows and rights of way. It also includes new sports pitches, equipped play areas and habitat creation in the form of new broadleaved planting, SUDS (and grassland meadows,

- Zone 3: Nightingale Park. This lies to the south east of the site and comprises sports pitches, allotments and habitat creation which include SUDS.

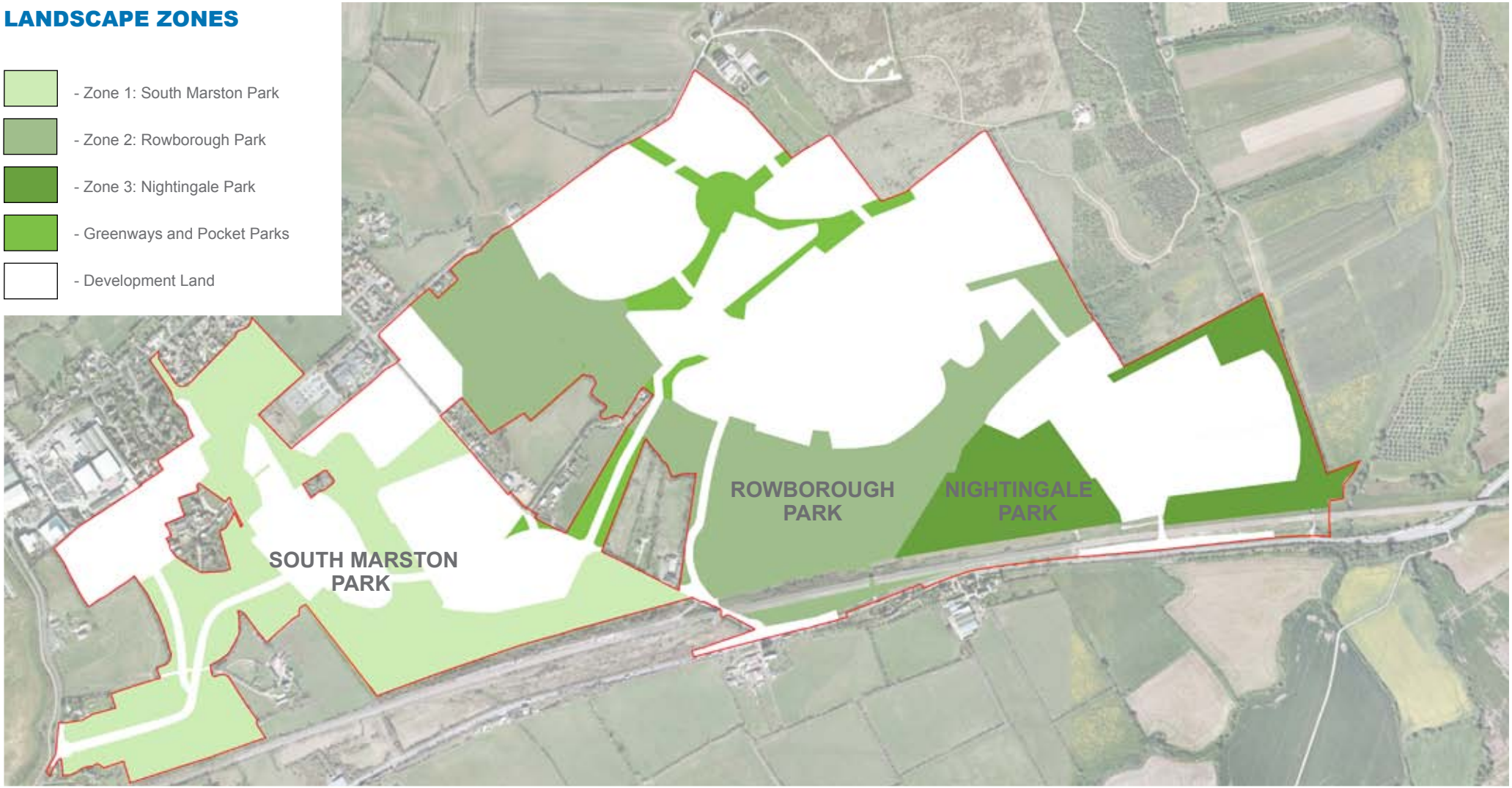
2) "Greenways" and "Pocket Parks". These are defined corridors or areas of natural greenspace that essentially connect with the three parks. They contain existing hedgerows and trees and deliver opportunities for habitat creation in the form of new planting, as well providing informal recreation to include walking, cycling and natural play.

4) Retained mature trees within the development parcels, which provide an opportunity in which to deliver feature spaces within the detailed layout.

5) New street trees and pockets of incidental greenspace and planting within development parcels, in addition to private gardens.

LANDSCAPE ZONES

- Zone 1: South Marston Park
- Zone 2: Rowborough Park
- Zone 3: Nightingale Park
- Greenways and Pocket Parks
- Development Land



ZONE 1
SOUTH MARSTON PARK

KEY FEATURES

- Existing Vegetation (Woodland, Hedges and Trees)
- Existing Public Footpath and Bridleway
- Proposed New Woodland, Hedgerow and Tree Planting
- Proposed Allotments
- Proposed Sports Pitches
- Proposed SUDs

* Indicative sketch

ZONE 2
ROWBOROUGH PARK

KEY FEATURES

- Existing Vegetation (Woodland, Hedges and Trees)
- Existing Public Footpath and Bridleway
- Proposed New Woodland, Hedgerow and Tree Planting
- Proposed Allotments
- Proposed Sports Pitches
- Proposed SUDs

* Indicative sketch

ZONE 3
NIGHTINGALE PARK

KEY FEATURES

- Existing Vegetation (Woodland, Hedges and Trees)
- Existing Public Footpath and Bridleway
- Proposed New Woodland, Hedgerow and Tree Planting
- Proposed Allotments
- Proposed Sports Pitches
- Proposed SUDs

* Indicative sketch

DESIGN PRINCIPLES

The Proposed Development would include a significant area of informal open space, thereby providing an opportunity to help the local population have access to and value a wildlife rich environment close to where they live and work. This is provided by an interlinking green infrastructure system having the South Marston Brook and Rowborough Park as its spine. Open land will be sustainably managed for people and wildlife; new built development will play a part in delivering new and enhanced habitats.

New hedgerows and scrub habitat would be created within public open space and within the GI, which provide a habitat corridor through the site and wider area, linking with off-site semi-natural habitats in the northeast and east. Shrub species should be based on the composition of existing hedgerows bounding and within the site. Where possible, existing valued hedgerows will be maintained as they provide sheltered corridors for wildlife to move along, encouraging dispersal throughout the site. Hedgerows already provide good structural diversity and associated under-storey vegetation adds more variety to the habitat. This may be sustained through appropriate management.

New woodland planting will be undertaken in order to augment and complement existing woodlands that exist locally such as Nightingale Wood to the north east, increase connectivity between them and buffer the existing woodlands from the effects of development. The planting mix would be representative of local woodlands. Little management would be undertaken initially to ensure that the area is as little disturbed as possible although some thinning/coppicing/ pollarding may be required in time to encourage the development of structurally diverse woodland.

Where new tree and shrub planting within the structural landscaping and GI is to be undertaken, native species will be used wherever possible to ensure the target for the creation of native woodland is met .

Species-rich areas of grassland will be located widely within the site in association with woodland and scrub habitats along greenways and more extensively in Rowborough Park, Nightingale Park, South Marston Park and built development areas, where intensive use and more formal management is not proposed.

It is also the intention to create a network of small glades and areas of more open grassland among areas of more developed habitat such as around the existing ponds in particular, which are linked by a network of greenways. This will provide the structural and microhabitat diversity required by a range of species including a number of the invertebrates, grass snake and scrub specialist birds known to be present locally or recorded within the Site, as well as amphibians including the great crested newt. Grassland habitats would be created in areas currently intensively managed for agriculture. Where possible, topsoils would be removed or mixed with subsoils to provide a substrate more conducive to the development of a more species-rich sward. Ideally and in more extensive areas of open space, a variation in topography would be created with shallow banks and undulations left to provide a high level of variation in micro-topography.

Coarse grassland areas to be created will also be allowed to develop on more developed soils to allow vigorous growth and the development of a structurally diverse habitat suitable for the colonisation of a range of species.

It is likely that the seeding of grassland habitats will be required in order to prevent the establishment of weed or other undesirable species. A neutral grassland seed mix of native provenance will therefore be used.

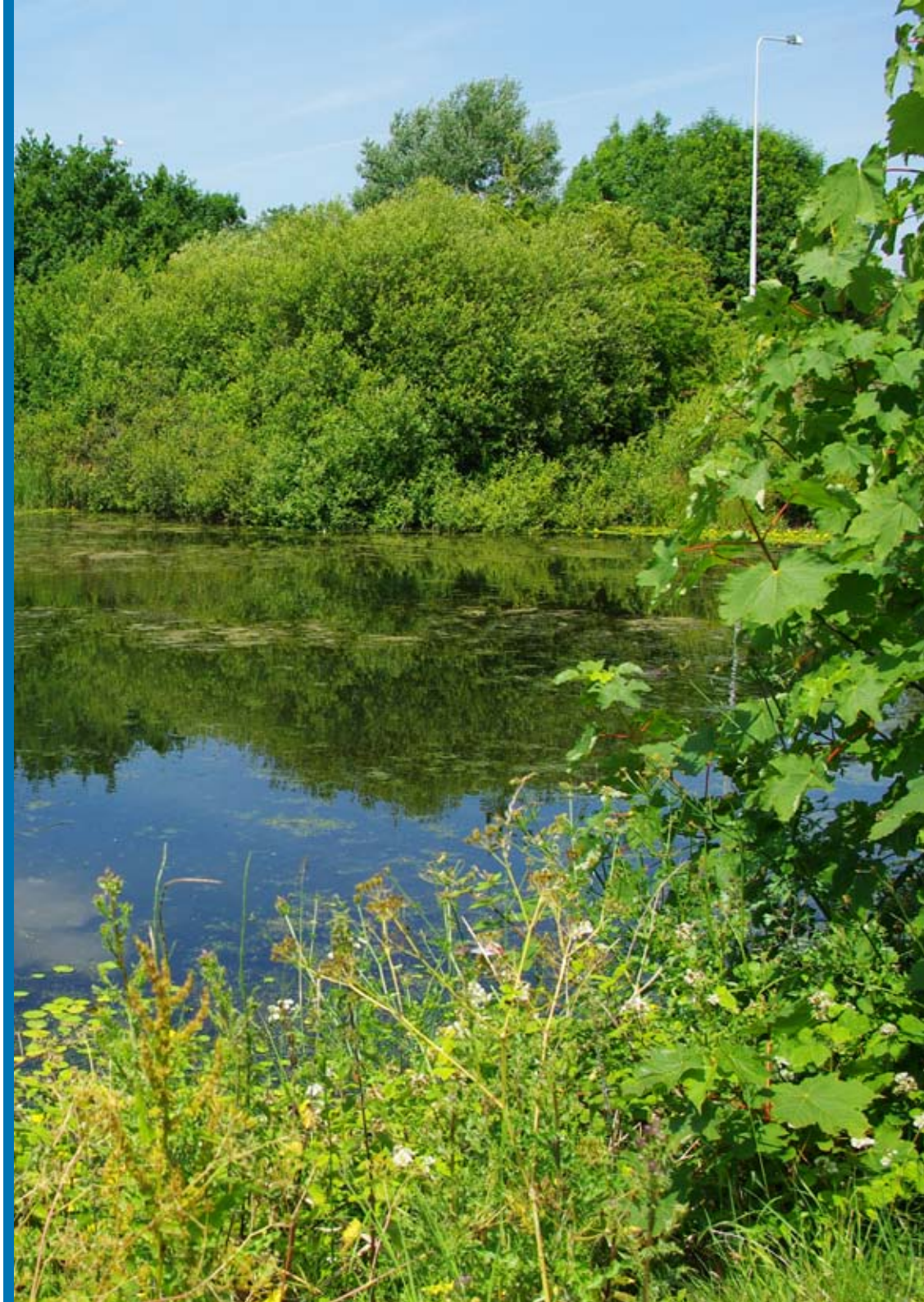
A variety of wetland features will be included within the site, both to benefit biodiversity but also to attenuate surface water flows.

Additional ponds will be formed close to the existing ponds to provide additional potential aquatic habitat for existing amphibian populations.

A significant opportunity also exists in association with surface water attenuation proposals to provide a range of habitats that would complement those of existing interest along the South Marston Brook. The primary function of the surface water drainage system will be to control flows and reduce the potential for pollution of the aquatic environment. However, It is important that SUDS are considered holistically and that each element is recognised as part of a network of habitats and wildlife corridors.

Principles followed will include:

- Where possible, SUDS will be located close to existing wetland areas, so plants and animals can naturally colonise the new SUDS ponds;
- Well vegetated shallow bays will be created allowing the establishment of areas of marsh;
- Smoothly finished surfaces would be avoided - although they give the impression of tidiness, they provide less micro habitat diversity for plants and animals; and
- If planting is essential, such as for the earlier phases, only native plants of local origin will be used.

SUMMARY

It is proposed that the detailed specification and programming of the GI Strategy proposals are undertaken in close consultation with relevant stakeholders.